Solid State Communications, Vol. 20, pp. 973-975, 1976. Pergamon Press. Printed in Great Britain.

OPTICAL ABSORPTION BY RARE-GAS ATOMS IN ALKALI-METAL RARE-GAS ALLOYS

R. C. Kittler and L. M. Falicov*
Department of Physics†
University of California, Berkeley
Berkeley, California 94720

(Received 30 August 1976 by A. A. Maradudin)

We present results of a Cluster-Bethe Lattice Method calculation of the optical absorption due to rare-gasses in alkali-metal rare-gas alloys. A specific application to the Cs-Xe alloy reveals that the main features of the absorption for the whole concentration range can be understood in terms of a single-quasiparticle model. The effects of short-range order are investigated.

The optical absorption by rare-gas atoms in alkali-metal hosts was recently measured by Flynn and coworkers[1-3] in the energy range 7 - 11 eV for concentrations ranging between 10^{-2} and 0.98 atomic fraction of rare gas. An absorption edge, related to the onset of the analogous $(p^6)\ ^1S_0 \rightarrow (p^5s)\ ^1P_1$ atomic transition, did not exhibit the singular behavior predicted by the many-body Hartree-Fock theory[4] for the dilute rare-gas limit. It was suggested that even in this limit the main features of the absorption might be explained in terms of the excitation of a single quasiparticle which is localized in the vicinity of the rare-gas atom. At concentrations of rare gas above a few percent the single quasiparticle view was more directly supported by the manner in which excitonic features, characteristic of rare-gas solids[5], became discernible in the absorption.

We present results of what we believe to be the first theoretical attempt based on the single quasiparticle model to explain the observed optical phenomena. Our approach neglects the broadening of the p-hole and assumes constant matrix elements between an Xe rare-gas core level and the 6s-band of Xe in Cs. With these simplifications the absorption is governed only by the local density of final s-states at the Xe atom and chemical shifts of the core level.

The local density of states (LDOS) at the Xe and Cs atoms were obtained using the Cluster-Bethe Lattice Method (CBLM)[6,7] applied to a tight-binding hamiltonian for a single s-orbital per site. Assuming a body-centered cubic lattice throughout the concentration range the LDOS were those of the central atoms of 9-atom clusters attached at their boundaries to an effective medium (Cayley tree)[7]. These LDOS were averaged over all possible cluster configurations consistent with the short-range order assumed. The input to the calculation consisted of the tight-binding parameters and the short-range order present.

In the spirit of the CBLM, the matrix elements of the hamiltonian were fitted to the known bandwidths and atomic data of the pure materials. With only nearest-neighbor interactions the hamiltonian can be written as:

$$H = \sum_i U_i |i\rangle\langle i| + \sum_{\langle ij \rangle} V_{ij} |i\rangle\langle j|$$

with site labels i and nearest-neighbor pairs $\langle ij \rangle$. The zero of the energy scale was taken to be the $^1S_0$ level of atomic Cs and the difference between the atomic levels of Cs and Xe was obtained from standard tables[8]. The unshifted core level of Xe was fitted to the position of the excitonic peak in the pure solid[5]. We assume a chemical shift of the on-site matrix elements which is linear in the number of Cs nearest neighbors of that site. The chemical shift at the Cs sites was derived from the cohesive and Fermi energies of metallic Cs[9], and at the Xe sites from the position of the absorption edge in the dilute Xe limit[1].

TABLE I

Tight-binding matrix elements used in the Cs-Xe calculation

matrix element	unshifted value (eV)	maximum chem. shift (eV)
U_{Cs}	0.	-0.20
U_{Xe}	0.20	0.40
V_{Xe-Xe}	0.	--
V_{Xe-Cs}	-0.20	--
V_{Cs-Cs}	-0.26	--
$E_{Xe\ CORE}$	-8.18	0.40

* On leave for the 1976-77 academic year at Cavendish Laboratory, Madingley Road, Cambridge CB3 OHE, England.

† Work supported in part by the National Science Foundation through grant DMR72-03106-A02.

The hopping matrix element between Cs atoms was chosen so as to reproduce the 6s bandwidth of pure Cs as determined from the bandstructure calculation of Mahanti and Das[10]. The Cs-Xe interatomic interaction was arbitrarily taken to be slightly less than the Cs-Cs term, primarily because it was expected that paths between Cs and Xe sites would be less mobile than those between two Cs sites. We found that a variation of the Cs-Xe term by up to ≈40% had little effect on the primary results of the calculation. The Xe-Xe interatomic parameter was taken to be zero so as to produce only localized excitons in pure Xe. Table I summarizes the tight-binding matrix elements used in the calculations.

The amount of short-range order present in the alloy is evidenced by two separate aspects of the experimental results. The height of the first main excitonic-like peak above the edge is linear in the concentration of rare-gas atoms as it would be in a random alloy if due to rare-gas pairs[2]. The d.c. resistivity of the Cs-Xe alloy[3] exhibits a smooth metal-to-insulator transition at a volume fraction of Xe only slightly higher than that obtained from numerical simulations of the percolation problem on random lattices[11]. In opposition to the tendency toward segregation which one might expect based on the contribution of the Cs conduction electrons to the cohesive energy, the absorption and resistivity results seem to

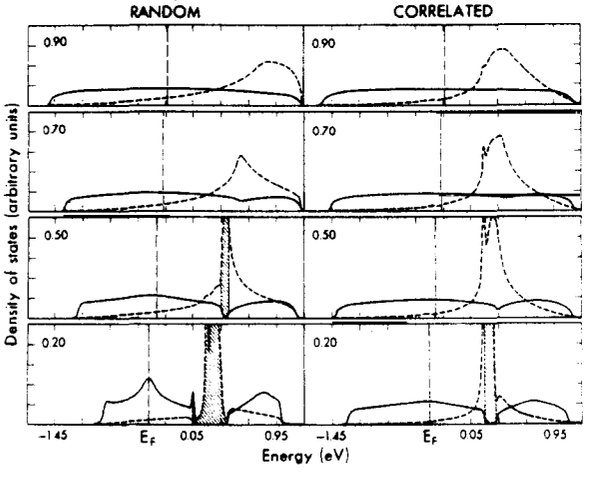


Fig. 1:
Calculated local densities of states on the Cs (full line) and Xe (dashed line) sites of the random and correlated Cs-Xe alloys. The concentration of Cs and the Fermi level are indicated. The localized states are denoted by the shaded areas.

indicate that there is no appreciable correlation present at all in this alloy. We have calculated the absorption and LDOS for the random correlation sequence and for a sequence representing a correlation halfway between the random and segregated alloys[7].

The cluster-averaged LDOS are shown in Fig. 1 for the two different correlation sequences at various concentrations of Xe in Cs. A small imaginary part has been added to the energy (0.015 eV) to broaden any delta-function (localized) states. The LDOS for very dilute concentrations of Xe in Cs are identical for the two sequences and not very different from those shown for $Cs_{0.9}Xe_{0.1}$. The LDOS of pure Xe consists of a single delta function near the main peak present in $Cs_{0.2}Xe_{0.8}$.

The main features of the alloy LDOS can be understood in terms of the differences in the atomic properties of Cs and Xe and the short-range order present. The cesium LDOS

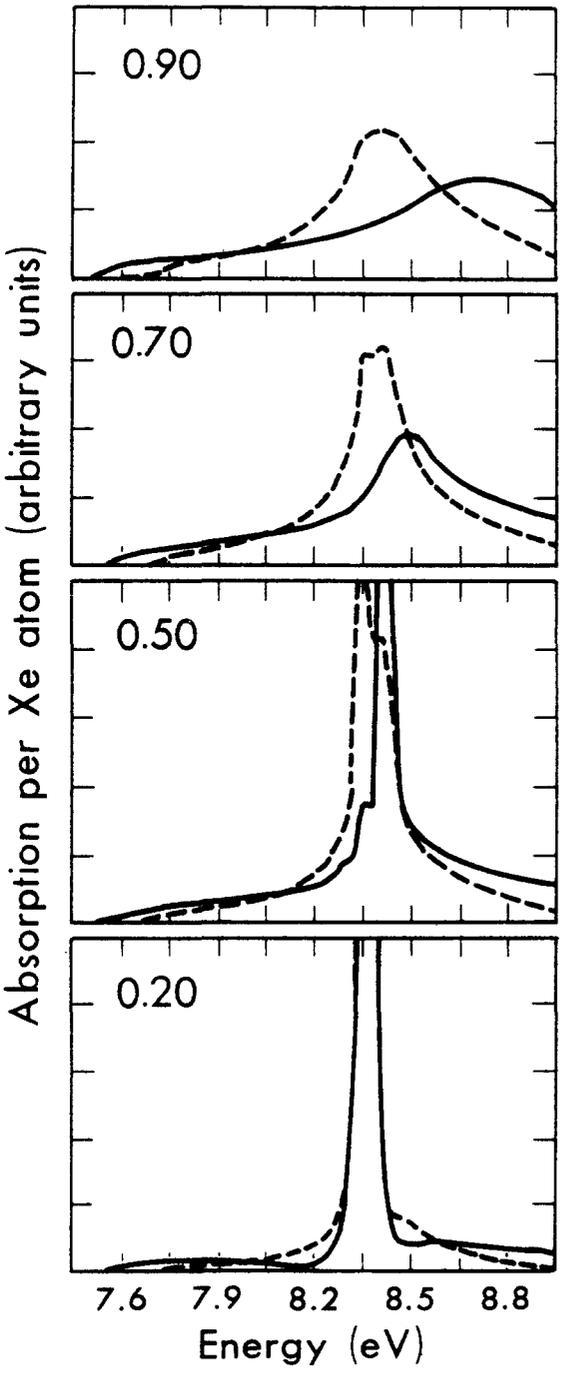


Fig. 2:
Absorption per Xe atom in the random (full line) and correlated (dashed line) Cs-Xe alloys. The concentration of Cs is indicated.

for $Cs_{0.9}Xe_{0.1}$ resembles that of the 6s-band of pure Cs. The broad peak present in the xenon LDOS at this concentration is due to the influence of Xe-Xe pair bonds in blocking itinerant paths and producing atomic-like Xe states in various cluster configurations. As the fraction of Xe increases these resonant peaks develop and move toward lower energy because of smaller chemical shifts. A gap in the band of itinerant Cs states appears when the concentration and correlation lead to such strong resonance conditions that the propagation along infinite paths is prevented. The tendency toward segregation results in this gap appearing later in the correlated sequence than in the random sequence even though the resonance peaks evolve more rapidly. Although the critical concentrations are undoubtedly tied to the effective medium we have chosen, it is reasonable to assume that the localization of states is inevitable in this particular alloy.

The position of the Fermi level changes only slightly upon alloying and in both sequences remains near the middle of the lower band, which arises primarily from Cs orbitals. As the concentration of Xe in Cs increases, this band narrows and its states become more localized due to increasing disorder. For very high concentrations of Xe there is a large probability of a Cs site being completely surrounded by Xe atoms and hence a peak begins to develop in the band at the bonding energy of Cs-Xe pairs. As this peak develops it pins the Fermi level. For only a few percent Cs in Xe the density of states has a three-peak structure. Two of the peaks are due to the bonding and antibonding states of the Cs-Xe pairs and between them there is the main peak at the atomic 6s level of Xe. The trends toward localization do not show evidence of the metal-to-insulator transition[3] observed in the d.c. resistivity at 73% Xe. It would require larger cluster calculations which more realistically treat the fluctuations of atomic arrangements and the connectivity of the lattice to determine the mobility edges responsible for this transition.

The absorption per Xe atom -- the local density of unoccupied states -- is shown in Fig. 2. There are similar trends in both the random and correlated sequences and these trends agree with those observed in the absorption of other alkali-metal rare-gas alloys[1-3]. As the concentration of Xe increases the peak grows and shifts toward lower energy, the edge shifts slightly toward higher energy and the characteristics of a continuum remain up to very high Xe concentrations. Due to the broadening inherent in the experimental curves and the simplicity of our model of the absorption, an accurate determination of the short-range order cannot be derived from the Figure. However, we would expect that, in the future, specific heat or Knight shift experiments could provide further insight based on the LDOS curves of Fig. 1.

In summary, the LDOS and absorption from the CBLM calculation corroborate the single quasiparticle interpretation of the optical properties of the rare-gas atom in alkali-metal rare-gas alloys. The model does not describe the threshold edge profile in the dilute Xe impurity limit very well. It succeeds however in interpreting the development of the main features of the absorption per rare-gas atom for the whole range of concentrations in terms of the LDOS on the rare-gas atom and the short-range order present.

Acknowledgements - The authors would like to thank Prof. K.D. Schotte and Dr. P. Schlottmann for helpful discussions.

REFERENCES

1. TILTON, R.A., PHELPS, D.J. and FLYNN, C.P., Phys. Rev. Lett. 32, 1006 (1974).
2. TILTON, R.A. and FLYNN, C.P., Phys. Rev. Lett. 34, 20 (1975).
3. PHELPS, D.J., AVCI, R. and FLYNN, C.P., Phys. Rev. Lett. 34, 23 (1975).
4. MAHAN, G.D., Solid State Phys. 29, 75 (1974).
5. BALDINI, G., Phys. Rev. 128, 1562 (1962).
6. JOANNOPOULOS, J. and YNDURAIN, F., Phys. Rev. B 10, 5164 (1974); YNDURAIN, F. and JOANNOPOULOS, J., Phys. Rev. B 11, 2957 (1975); YNDURAIN, F. and FALICOV, L.M., Solid State Commun. 17, 1545 (1975); FALICOV, L.M. and YNDURAIN, F., Phys. Rev. B 12, 5664 (1975).
7. KITTLER, R.C. and FALICOV, L.M., J. Phys. C., to appear.
8. MOORE, C.E., Atomic Energy Levels-III, U.S. Gov't. Printing Office, Washington, D.C. (1958).
9. KITTEL, C., Introduction to Solid State Physics, 4th edition, pp.96, 248 Wiley, New York (1971).
10. MAHANTI, S.D. and DAS, T.P., Phys. Rev. 183, 674 (1969).
11. SCHER, H. and ZALLEN, R., J. Chem. Phys. 53, 3759 (1970).